INCH-POUND
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DETAIL SPECIFICATION

Cartridge, 105MM, HE, M1
Loading, Assembling and Packing

Inactive for new design after 8 February 1996

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the loading, assembling and packing of Projectile, 105MM, HE, M1 for use with the 105MM Howitzer M101, M102 and M119(see 6.1).

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardecstdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents sited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition lot numbering and ammunition data cards

MIL-STD-1916 - DoD preferred methods for acceptance of product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9211610 – Projectile, 105MM, HE, M1, Loading Assembly

9211611 – Cartridge, 105MM, HE, M1, w/o Fuze for Howitzer M2A1, M2A2, M103 and M137 Assembly

8838130 – Primer, Percussion, M28B2 Assembly

8838129 – Primer, Percussion, M28A2 Assembly

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8595386 – Case, Cartridge 105MM, M14B4 Assembly
7548409 – Case, Cartridge 105MM, M14 Assembly

7548025 – Case, Cartridge 105MM, M14B1 Assembly

9205472 – Charge, Propelling,M67 Assembly

7549073 – Container, Ammo, Fiber, M105A3/105MM Howitzer

7549072 – Box, Packing, Ammunition, for 105MM Howitzer Ammunition in Fiber Container M105A3
8802117 – Testing Fixture Primer Sensitivity Assembly

(Copies of these drawings may be requested online at Pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM E 1742 - Standard Practice for Radiographic Examination

(Copies of ASTM standards are available online at http://www.astm.org or from American Society for Testing and materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Components and assemblies.

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3.2.1 Projectile loading assembly (prior to loading explosive). The components and assemblies shall comply with all requirements specified on dwg. 9211610.

3.2.2 Projectile loading assembly (Radiographic acceptance requirements). The components and assemblies shall comply with all requirements specified on dwg. 9211610.

3.2.3 Projectile loading assembly (prior to assembly of liner). The components and assemblies shall comply with all requirements specified on dwg. 9211610.

3.2.4 Projectile, 105MM, HE, M1 LAP (prior to assembly of supplementary charge). The components and assemblies shall comply with all requirements specified on dwg. 9211610.

3.2.5 Projectile 105MM, HE, M1 LAP (prior to assembly of closing plug). The components and assemblies shall comply with all requirements specified on dwg. 9211610.

3.2.6 Primer (prior to assembly to case). The components and assemblies shall comply with all requirements specified on dwg. 8838130 or 8838129.

3.2.7 Cartridge case (prior to assembling propelling charge). The components and assemblies shall comply with all requirements specified on dwg. 8595386, 7548409 or 7548025.

3.2.8 Propelling charge (prior to assembly into cartridge case). The components and assemblies shall comply with all requirements specified on dwg. 9205472.

3.2.9 Cartridge case with propelling charge. The components and assemblies shall comply with all requirements specified on dwg. 9211611

3.2.10 Projectile and cartridge case assemblies (prior to pack). The components and assemblies shall comply with all requirements specified on dwg. 9211611

3.2.11 Unsealed fiber container. The components and assemblies shall comply with all requirements specified on dwg. 7549073.

3.2.12 Sealed fiber container. The components and assemblies shall comply with all requirements specified on dwg. 7549073.

3.2.13 Sealed wood packaging box. The components and assemblies shall comply with all requirements specified on dwg. 7549072.

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3.2.14 Ballistic test. The components and assemblies shall comply with all requirements specified on dwg. 9211611.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Loading of high explosive

3.4.1 Interior cavity metal defects. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, embedded foreign matter, folds, pits and sharp edges, in accordance with visual standards (see 6.3).

3.4.2 Unsolidified pool of paint. There shall be no unsolidified pool of paint or blister in the projectile cavity prior to loading of the explosive. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile.

3.4.3 Projectile cavity contamination. The interior cavity of the projectile prior to loading shall be free from foreign matter, chemical residue, oil and grease.

3.4.4 Interior coating. Projectile having the interior wall imperfectly coated, in accordance with visual standards (see 6.3) shall not be loaded.

3.4.5 Solid explosive. Solid explosive shall not be present in the projectile prior to loading or be added to the shell during pouring or cooling.

3.5 Radiographic inspection.

3.5.1 Metal parts defects. There shall be no cracks, pipes or thin wall sections in the projectile metal part. There shall be no evidence of gaps, as defined by visual x-ray standards, between the rotating band and its seat (see 6.3.)

3.5.2 Foreign material. There shall be no foreign material in the explosive charge.

3.5.3 Base separation. There shall be no visible base separation between the explosive charge and the shell base.

3.5.4 Porosity (see 6.17.)

3.5.4.1 Shrink porosity. Shrink porosity within the explosive charge shall not exceed the requirements specified in Table I.

3.5.4.2 Spherical porosity. Spherical voids greater than 1/8 inch in diameter shall be considered as cavities. Spherical

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pores less than 1/32 inch in diameter shall be disregarded. Visual x-ray image standards illustrating acceptable levels of spherical porosity shall be established for each line and process (see 6.3). Areas of spherical porosity not exceeding the standards shall be acceptable.

3.5.5 Projected cavities. Cavities within the explosive charge shall not exceed the requirements specified in Table I. Cavities having a maximum projected length of 1/32 inch shall be disregarded.

3.5.6 Piping (also known as core porosity). Piping within the explosive charge shall not exceed the requirements specified in Table I.

3.5.7 Cracked charges.

3.5.7.1 Width. The width of a transverse crack shall be measured within 1/8 inch of where it intersects the projectile body. If the width is not clearly measurable within 1/8 inch of the body intersection or it does not extend to within 1/8 inch of the body intersection, that transverse crack shall be considered to have met the width requirement of Table I. The plane of a transverse crack shall appear to intersect both projectile sidewalls, to include the base radius. Transverse cracks which meet the width requirement shall be counted for the sum requirement. All cracks, regardless of orientation, shall meet the width requirement.

3.5.7.2 Sum. Not more than one transverse crack shall be permitted in Segment A and not more than two transverse cracks shall be permitted n Segment A, B and C combined. Cracks which appear to be joined at any point shall be considered as one crack. Transverse cracks on x-ray film or display appear as a group or cluster of lines or elements because the crack in the explosive is fragmented along various fracture lines. An apparent collection of such transverse elements that are joined together shall be considered as one crack. Separated transverse cracks apparently joined by a longitudinal crack, know as “Christmas tree” cracks shall be counted as separate cracks.

3.5.8 Annular ring. An annular ring within the explosive charge shall not exceed the requirements specified in Table I.

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TABLE I. Radiographic defects size

Cast Defects		Segments (See figure 1)			
		A	B	C	D
Sum of projected areas of shrink porosity	(in^2)	5/16	5/8	5/8	-
Sum of projected areas of cavities, excluding pipes, cracks, annular rings and porosity.	(in^2)	*1/64	1/4	1/2	1/2
Projected length of any cavity excluding pipes, cracks, annular rings and porosity.	(in)	1/8	1/2	1/2	3/4
Piping cavities maximum projected area.	(in^2)	0	1/4	1/2	-
Piping cavities maximum projected width.	(in)	0	1/4	1/4	-
Cracks, maximum projected width.	(in)	1/32	1/32	1/16	-
Annular rings, maximum projected width.	(in)	0	0	1/4	-
Notes:					
* If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 in^2.					
Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis three times or greater than the width.					
An Annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.					

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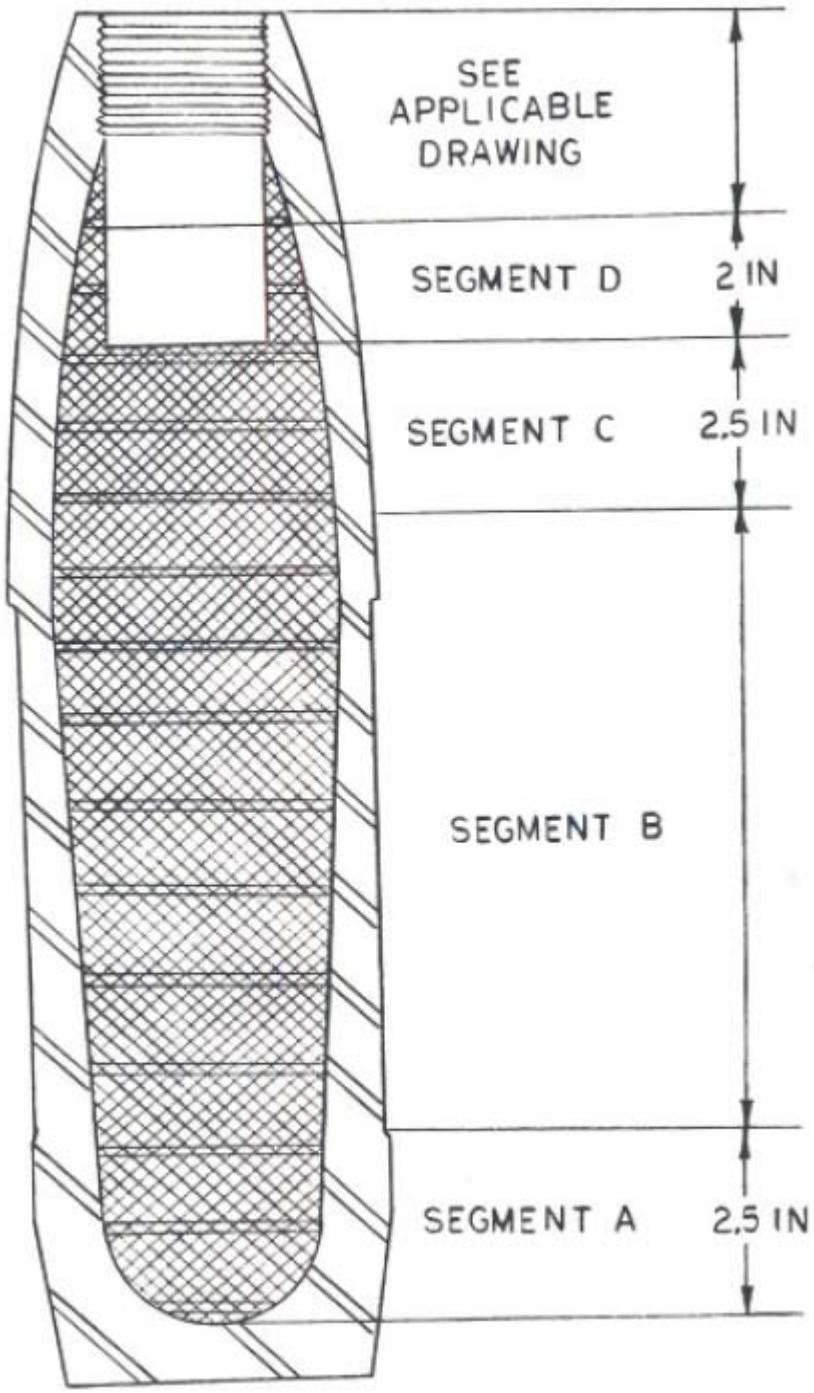


Figure 1. Projectile, HE, 105MM, M1 Explosive Cast Segments

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3.6 Mating thread surfaces. The mating surfaces of threads shall be free of caked explosive and free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care should be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.7 Fuze well surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The maximum dimension of any visible cavity in the faced surface of the charge shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable region of the fuze well prescribed by 9211610.)

3.8 Base separation saw samples. The maximum allowable base separation shall be .015 inch when inspected after sectioning.

3.9 Fuze well (before assembly of the liner). The fuze well shall be free from any loose explosive or any foreign matter prior to assembly of the liner. Chemical cleaner shall not contact the charge or become entrapped in the projectile.

3.10 Fuze well (after assembly of the liner). The fuze well shall be free from any loose explosive or any foreign matter after assembly of the liner.

3.11 Exterior surfaces explosive contamination. There shall be no stain, TNT bloom or caked explosive (see 6.6) on the exterior of the projectile prior to shipping from the load plant.

3.12 Ballistic test.

3.12.1 Metal parts breakup. There shall be no premature burst or indication of metal parts separation in the gun or in flight.

3.12.2 Shearing of rotating band. There shall be no shearing of the rotating band as evidenced by complete lack of engraving.

3.12.3 Rotating band failure. There shall be no loss or fanning (any lifting of rotating band edge) of the rotating band.

3.12.4 Muzzle Velocity. The muzzle velocity for the cartridge, fired at 70 +/-2.5 degrees Fahrenheit, shall not be

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more than 1642 feet/second nor less than 1592 feet/second (corrected for a standard 33lb fuzed projectile) and the standard deviation of the lot shall not be greater than 8 feet per second. The value of k_U or k_L (Acceptable Quality Levels, see 4.4.2.1) shall be greater than or equal to 1.85 for the initial sample and the value of k_U or k_L shall be greater than or equal to 1.72 for the reduced sample.

3.13 Workmanship. The requirements of workmanship are as follows:

a. Cleaning. If required the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

b. Painting. Painting shall comply with the requirements of dwg. 9211610. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

c. Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.13.1 Rotating band. The land surfaces of the rotating band shall comply with the following:

a. There shall be no raised brass.

b. There shall be no paint or foreign matter on the outer diameter.

c. Nicks, gouges, scratches and other depressions on the lands not exceeding 0.025 inches in depth are acceptable.

d. Nicks, gouges, scratches and other depressions on the lands with depths greater than 0.025 and less than or equal to 0.050 inches are acceptable unless damage extends longitudinally across more than half of the land width.

e. There shall be no “washboard” or striated area that extends longitudinally across more than half of the land width if the depth exceeds 0.025 inches at any point.

f. There shall be no nicks, gouges, scratches or other depressions on the lands with depths greater than 0.050 inches.

3.14 Inspection lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916.

3.15 Deterioration of primer, percussion, M28. If the total elapsed time between original acceptance of the M28

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primer lot and the assembly of that lot into cartridges exceeds three years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in 4.4.4, immediately before the M28 primer lot is assembled into cartridges (see 6.21).

3.15.1 Non-functioning. The head assembly shall not function by an applied energy of 47.52 inch-ounces minimum (min).

3.15.2 Functioning. The head assembly shall function by applied energy of 159.20 inch-ounces maximum (max).

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4. VERIFICATION

TABLE II. Requirement/verification cross reference matrix

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 - Test							Classes of Verification A- First Article B- Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1.1	First article				X	X	X		4.2
3.1.2	Conformance				X	X		X	4.3
3.2.1	Projectile loading assembly (prior to loading explosive)				X	X	X	X	4.3.2.1
3.2.2	Projectile loading assembly (Before post cycle heating)(Radiographic acceptance requirements)				X	X	X	X	4.3.2.2
3.2.3	Projectile loading assembly (prior to assembly of liner)				X	X	X	X	4.3.2.3
3.2.4	Projectile, 105MM, HE, M1 LAP (prior to assembly of supplementary charge, if applicable)				X	X	X	X	4.3.2.4
3.2.5	Projectile 105MM, HE, M1 LAP (prior to assembly of closing plug)				X	X	X	X	4.3.2.5
3.2.6	Primer (Prior to assembly to case)				X	X	X	X	4.3.2.6
3.2.7	Cartridge case (prior to assembling propelling charge)				X	X	X	X	4.3.2.7
3.2.8	Propelling charge (prior to assembly into cartridge case)				X	X	X	X	4.3.2.8

TABLE II. Requirement/Verification Cross Reference Matrix- Continued

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3.2.9	Cartridge case with propelling charge				X	X	X	X	4.3.2.9
3.2.10	Projectile and cartridge case assemblies (prior to pack)				X	X	X	X	4.3.2.10
3.2.11	Unsealed fiber container				X	X	X	X	4.3.2.11
3.2.12	Sealed fiber container				X	X	X	X	4.3.2.12
3.2.13	Sealed wood packaging box				X	X	X	X	4.3.2.13
3.2.14	Ballistic test				X	X	X	X	4.3.2.14
3.3	Ammunition lot numbering				X		X	X	4.3.1
3.4.1	Interior cavity metal defects				X	X	X	X	4.3.2.1
3.4.2	Unsolidified pool of paint.				X	X	X	X	4.3.2.1
3.4.3	Projectile cavity contamination				X	X	X	X	4.3.2.1
3.4.4	Interior coating				X	X	X	X	4.3.2.1
3.4.5	Solid explosive				X	X	X	X	4.3.2.1
3.5.1-3.5.8	Radiographic Inspection				X	X	X	X	4.3.2.2/4.4.3
3.6	Mating surfaces				X	X	X	X	4.3.2.3
3.7	Fuze well surface cavitation				X	X	X	X	4.3.2.3
3.8	Base separation saw samples				X	X	X	X	4.3.2.3/4.4.1
3.9	Fuze well (before)				X	X	X	X	4.3.2.3
3.10	Fuze well (after assembly of the liner)				X	X	X	X	4.3.2.4
3.11	Exterior surfaces explosive contamination				X	X	X	X	4.3.2.4
3.12.1	Metal parts breakup					X	X	X	4.3.2.14/ 4.4.2
3.12.2	Shearing of rotating band					X	X	X	4.3.2.14/ 4.4.2
3.12.3	Rotating band failure					X	X	X	4.3.2.14/ 4.4.2
3.12.4	Muzzle velocity					X	X	X	4.3.2.14/ 4.4.2/4.4.2.1
3.13	Workmanship				X		X	X	4.3.2.1 thru 4.3.2.13

TABLE II. Requirement/Verification Cross Reference Matrix- Continued

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3.13.1	Rotating Band				X	X	X	X	4.3.2.5
3.14	Inspection lot formation				X			X	4.3.1
3.15.1	Non-functioning				X	X	X	X	4.3.2.6/ 4.4.4.1
3.15.2	Functioning				X	X	X	X	4.3.2.6/ 4.4.4.2

4.1 Classification of inspections. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with Table III.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table III.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table III.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE III First article inspection

	Number of FAT Samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Projectile loading assembly (dwg 9211610)			
Examination for defects, Note 1	10	3.2	4.3.2.1-4.3.2.5
Base separation test	10	3.8	4.4.1
Cartridge, 105mm, HE, M1 Loading, Assembly and Marking (Dwg 9211611)			
Examination for defects, Note 1	30	3.2	4.3.2.6-4.3.2.13
Ballistic test (Safety)	25	3.12	4.4.2.2
Ballistic test (Muzzle velocity)	15	3.12	4.4.2.1

Note 1. The examination for defects in paragraph 4.3.2 are for dimensional and visual inspections and in many cases must be verified during assembly.

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916. In addition, each inspection lot of the cartridge 105MM HE M1 shall contain:

a. Explosive of one type from one lot interfix number from one manufacturer.

b. Projectile metal parts of one lot interfix number from one manufacturer.

c. Loaded projectiles of one weight zone.

d. Supplementary charge of one lot interfix number from one manufacturer.

e. Cartridge cases of one lot interfix number from one manufacturer.

f. Propelling charges from not more than one lot.

g. Loaded primers of one lot interfix number from one manufacturer.

h. Fuze well liners of one lot interfix number from one manufacturer.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 through 4.3.2.14. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative Inspections. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916.

d. Tool control. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Projectile loading assembly (prior to loading explosive)	SHEET 1 OF 1		DRAWING NUMBER 9211610 / NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Foreign matter in charge cavity	100%	3.4.3	Visual/ Inspection light
Major				
101	Interior protective coating with bare spot	100%	3.4.4	Visual Standard/ Inspection light
102	Pool of paint in bottom of cavity	100%	3.4.2	Visual
103	Metal defects in cavity	100%	3.4.1	Visual/ Inspection light
104	Solid explosive present	100%	3.4.5	Visual/ Inspection light
Minor				
201	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Projectile loading assembly (Before post cycle heating) (Radiographic acceptance requirements)	SHEET 1 OF 3	DRAWING NUMBER 9211610 / NEXT HIGHER ASSEMBLY 9211611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Base separation in excess of 0.015" (see 6.16.1)	100%	3.5.3	4.4.3
2	Cavities, other than piping, cracks or annular rings in Segment A with a total area in excess of 1/16 square inch or length in excess of 1/4 inch (see 6.16.2)	100%	3.5.5	4.4.3
3	Annular rings in Segment A or B with a width in excess of 1/32 inch (see 6.16.3)	100%	3.5.8	4.4.3
4	Piping cavities in Segment A exceeding 1/16 square inch in total area or 1/8 inch in width (see 6.16.4)	100%	3.5.6	4.4.3
5	Cracks in Segment A in excess of the requirements of Table I (see 6.16.5)	100%	3.5.7.1	4.4.3

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Projectile loading assembly (Before post cycle heating) (Radiographic acceptance requirements)	SHEET 2 OF 3		DRAWING NUMBER 9211610 / NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
6	Cast defects in Segment B, except for porosity in excess of twice the requirements of Table I (see 6.16.6)	100%	3.5.4 through 3.5.8	4.4.3
7	Metal parts defects, except rotating band gaps, in excess of that permitted (see 6.11 and 6.16.7)	100%	3.5.1	4.4.3
8	Foreign material in the explosive cast (see 6.16.8)	100%	3.5.2	4.4.3

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Projectile loading assembly (Before post cycle heating) (Radiographic acceptance requirements)	SHEET 3 OF 3		DRAWING NUMBER 9211610 / NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Major				
101	Cast defects in Segments A (except cracks,) B, C or D in excess of the requirements of table 1	100%	3.5.4 through 3.5.8	4.4.3
102	Porous areas in excess of that permitted	100%	3.5.4	4.4.3
103	Cracks in Segment A in excess of that permitted	100%	3.5.7.2	4.4.3
104	Rotating band gaps in excess of that permitted	100%	3.5.1	4.4.3
105	Any visible base separation, less than 0.015”	100%	3.5.3	4.4.3
Minor				
201	Cracks in Segments A, B and C combined in excess of that permitted	100%	3.5.7.2	4.4.3

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Projectile loading assembly (prior to assembly of liner)	SHEET 1 OF 1	DRAWING NUMBER 9211610
			NEXT HIGHER ASSEMBLY 9211611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Base separation, destructive test see(6.16.9)	4.4.1	3.8	4.4.1.1
Major				
101	Fuze well surface cavitation excessive	Level IV	3.7	Visual/ Dedicated Illumination(DI)
102	Loose explosive in fuze well	Level IV	3.9	Visual/DI
103	Foreign material in fuze well	Level IV	3.9	Visual/DI
104	Depth of fuze well.	Level IV	3.2.3	AIE
105	Diameter of fuze well max.	Level IV	3.2.3	AIE
106	Projectile threads not clean prior to application of adhesive	Level IV	3.2.3	Visual/DI
107	Caked explosive on mating surfaces of threads	100%	3.6	AIE
108	Application of adhesive incomplete or incorrect	Level IV	3.2.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Projectile, 105MM, HE, M1 LAP (prior to assembly of supplementary charge)	SHEET 1 OF 1	DRAWING NUMBER 9211610
			NEXT HIGHER ASSEMBLY 9211611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Depth to bottom of liner min.	Level IV	3.2.4	AIE
102	True position of inside diameter of liner with minor diameter of internal thread and end face of projectile body	Level IV	3.2.4	AIE
103	Inside diameter of liner after swaging	Level IV	3.2.4	AIE
104	Torque of liner (witnessed during application of assembly torque)	Level IV	3.2.4	AIE/Visual
105	Swage missing or incomplete	100%	3.2.4	Visual
106	Presence of caked explosive, stain or bloom on exterior of shell body, and forward face	100%	3.11	Visual
Minor				
201	Liner damaged	Level II	3.2.4	Visual
202	Foreign material in liner cavity including explosive	Level II	3.10	Visual
203	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Projectile 105MM, HE, M1 LAP (prior to assembly of closing plug)	SHEET 1 OF 2		DRAWING NUMBER 9211610 / NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Supplementary charge pad or supplementary charge missing, (see 6.16.11)	100%	3.2.5	AIE
2	Projectile zone weight incorrect by two or more zones	100%	3.2.5	AIE/Balance
3	Zone weight stenciling or punch marks incorrect by two or more zones	100%	3.2.5	Visual/Manual
Major				
101	Diameter of bourrelet, max. (applies if bourrelet has been retouched)	100%	3.2.5	AIE
102	Supplementary charge inverted	Level IV	3.2.5	Visual
103	Zone weight stenciling or punch marks incorrect	Level IV	3.2.5	AIE/Balance
104	Supplementary charge not removable by hand	Level IV	3.2.5	Manual
105	Projectile zone weight incorrect	100%	3.2	AIE/Balance

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Projectile 105MM, HE, M1 LAP (prior to assembly of closing plug)	SHEET 2 OF 2	DRAWING NUMBER 9211610
			NEXT HIGHER ASSEMBLY 9211611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Marking missing, misleading or unidentifiable	Level II	3.2.5	Manual
202	Supplementary charge damaged	Level II	3.2.5	Visual
203	Foreign matter on threads including explosive or exudates	Level II	3.6	Visual
204	Spacer missing or damaged	Level II	3.2.5	Visual
205	Rotating band damaged	Level II	3.13.1	Visual
206	Evidence of poor workmanship	Level II	3.13	Visual
207	Protective coating with total damaged area in excess of ¼ square inch	Level II	3.2.5	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	**TITLE** Primer (prior to assembly to case)		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 8838130 or 8838129
				NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	One or more flash holes missing	100%	3.2.6	Visual
Major				
101	Non-functioning check test of primer	4.4.4.3	3.15.1	4.4.4.1
102	Functioning check test of primer	4.4.4.3	3.15.2	4.4.4.2
103	Primer assembly defective	100%	3.2.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.7	**TITLE** Cartridge case (prior to assembling propelling charge)	**SHEET** 1 **OF** 1		**DRAWING NUMBER** 8595386, 7548409 or 7548025
				NEXT HIGHER ASSEMBLY 9211611
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Primer above flush	100%	3.2.7	AIE
Major				
101	Primer missing or damaged	Level IV	3.2.7	Visual
Minor				
201	Foreign matter in interior of case	Level II	3.13	Visual
202	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8	TITLE Propelling charge (prior to assembly to cartridge case)	SHEET 1 OF 1	DRAWING NUMBER 9205472
			NEXT HIGHER ASSEMBLY 9211611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Increment missing	100%	3.2.8	AIE
Major				
101	Any opening in bag large enough to permit escape of propellant	Level IV	3.2.8	Visual
Minor				
201	Cord broken (M67 charge)	Level II	3.2.8	Visual
202	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.9	TITLE Cartridge case with propelling charge (prior to assembling to projectile)	SHEET 1 OF 1	DRAWING NUMBER 9211611 / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical 1	Incorrect sequence of increments	100%	3.2.9	AIE
Major 101	Foil side of bag, charge 5, not facing primer	Level IV	3.2.9	Visual
Minor 201	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.10	TITLE Cartridge assembly	SHEET 1 OF 1	DRAWING NUMBER 9211611
			NEXT HIGHER ASSEMBLY 7549073

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Torque test failure of plug	Level IV	3.2.10	AIE
102	Any component missing	Level IV	3.2.10	Visual
Minor				
201	Exterior coating with total damaged area in excess of ¼ square inch	Level II	3.2.10	Visual
202	Marking missing, misleading or unidentifiable including incorrect color (except zone weight)	Level II	3.2.10/3.3	Visual
203	Plug not fully seated	Level II	3.2.10	Visual
204	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.11	TITLE Unsealed fiber container	SHEET 1 OF 1	DRAWING NUMBER 7549073
			NEXT HIGHER ASSEMBLY 7549072

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Packing component missing, damaged or improperly assembled	Level IV	3.2.11	Visual
102	Cartridge cannot be inserted or removed by hand	Level IV	3.2.11	Manual
103	Container covers cannot be removed by hand	Level IV	3.2.11	Manual
104	Insufficient fillers for tight pack	Level IV	3.2.11	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.13	Visual/Manual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.12	TITLE Sealed fiber container	SHEET 1 OF 1	DRAWING NUMBER 7549073
			NEXT HIGHER ASSEMBLY 7549072

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealing strip incomplete	Level IV	3.2.12	Visual
102	Sealing strip wrinkled	Level IV	3.2.12	Visual
103	Sealing strip material improper	Level IV, Note 1	3.2.12	Visual/Certification
104	Sealing strip damaged	Level IV	3.2.12	Visual
105	Interior fiber moisture content excessive at time of pack	Level IV	3.2.12	AIE
106	Component missing	Level IV	3.2.12	Visual
107	Marking misleading, missing, incorrect, or illegible	Level IV	3.2.12	Visual
108	Projectile not packed tightly	Level IV	3.2.12	Manual/Audible
Minor				
201	Tear tab length below minimum	Level II	3.2.12	AIE
202	Components damaged	Level II	3.2.12	Visual
203	Evidence of poor workmanship	Level II	3.13	Visual

Note 1, 1 certification per batch

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.13	TITLE Sealed wood packing box	SHEET 1 OF 1	DRAWING NUMBER 7549072
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Box damaged or exposing contents	Level IV	3.1	Visual
102	Hardware or strapping missing, broken or loose	Level IV	3.1	Visual
103	Board broken	Level IV	3.1	Visual
104	Handle missing or insecure	Level IV	3.1	Visual/Manual
Minor				
201	Strapping or hardware improperly assembled	Level II	3.1	Visual/Manual
203	Metallic seal missing, unsealed or improperly positioned	Level II	3.1	Visual
204	Marking missing, misleading or unidentifiable	Level II	3.1	Visual
205	Evidence of poor workmanship	Level II	3.13	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.14	TITLE Ballistic Test	SHEET 1 OF 1	DRAWING NUMBER 9211611
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Premature burst or metal parts breakup	4.4.2.1, 4.4.2.2	3.12.1	4.4.2
2	Shearing of rotating band	4.4.2.1, 4.4.2.2	3.12.2	4.4.2
3	Rotating band failure	4.4.2.1, 4.4.2.2	3.12.3	4.4.2
Major				
101	Muzzle Velocity	4.4.2.1	3.12.4	4.4.2

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4.4 Testing.

4.4.1 Base separation destructive test samples. Any projectiles with *visible* base separation less than 0.015" on x-ray shall be selected for this test. If insufficient samples are available with base separation, one sample shall be randomly selected per shift. If any sample projectile when tested has base separation in excess of the requirements of 3.8, the projectiles represented by the sample shall be rejected. The test shall be performed as specified in 4.4.1.1 (see 6.16.9). Sectioning of samples shall cease after one excessive gap is detected. For disposition of rejected lots, see 6.2.1.

4.4.1.1 Base separation destructive test. This test shall be conducted after post cycle heat treatment is completed. The temperature of the sample projectiles (metal parts and explosive) shall be 70±9 degrees F when sectioned and tested for base separation. To section the projectile cut parallel to the base approximately 2.25 inches up from the base to the center of the projectile axis, next cut through the base toward the nose perpendicular to aforementioned cut so as to remove a wedge of steel to expose explosive and metal parts interface. Nose drop the projectile with a current US 155MM HE artillery lifting plug assembled 1.5 inches onto steel plate of 1 inch minimum thickness and measure for base separation with a .5 inch wide feeler gage. Sectioning of saw samples shall cease after one excessive gap is detected.

4.4.2 Ballistic test. This test will be performed at a proving ground in accordance with 4.4.2.1 and 4.4.2.2. The samples selected will be fired in an M20 series cannon with muzzle break with acceptable tube life remaining. The projectiles will be fuzed with an M739 series fuze. Adequate photographic and instrumental coverage will be provided to assure compliance with 3.12. If cartridges are received with "ballistic sample numbers" from the contractor then the cartridges will be randomized with regard to test phase and order of firing. The following information pertaining to this test will be obtained and reported for all phases as specified below. For issues on test validity, see 6.2.

a. Photographic coverage. Photographic coverage will be adequate to assure compliance with 3.12.

b. Radar coverage. Doppler radar will be used.

c. Pressure. Pressure will be determined by the use of copper crusher gages.

4.4.2.1 Muzzle velocity.

4.4.2.1.1 First article and initial production lot samples. Beginning with the first article and first lot produced and

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continuing until three consecutive lots have complied with all the acceptance criteria contained herein, 25 sample projectiles shall be selected at random from each lot for this test. Compute the average muzzle velocity ($\bar{x}$) and standard deviation (s) using the (n-1) formula. Compute k_U = (1642 – $\bar{x}$)/s, and k_L = ($\bar{x}$-1592)/s. Reject the lot if either k_U or k_L is less than 1.85, if $k_U + k_L < 4.28$ or if any test cartridge fails to meet the requirements of paragraph 3.12. In addition if the standard deviation of the sample projectiles multiplied by a factor of 0.75 exceeds the lot requirement, the lot shall be rejected.

4.4.2.1.2 Production lot samples. Once three consecutive lots have met the criteria of 4.4.2.1.1, 10 projectiles shall be selected at random from each lot for this test. Compute the average muzzle velocity ($\bar{x}$) and the standard deviation(s) using the (n-1) formula. Compute k_u = (1642-$\bar{x}$)/s, and k_L = ($\bar{x}$-1592)/s. Reject the lot if either k_U or k_L is less than 1.72, if $k_U + k_L < 3.74$ or if any test cartridge fails to meet the requirements of paragraph 3.12. In addition if the standard deviation of the sample projectiles multiplied by a factor of 0.65 exceeds the lot requirement, the lot shall be rejected. In the event that any lot fails to meet all the requirements of this paragraph, successive lots must meet the provisions of 4.4.2.1.1 and the requirements specified therein.

4.4.2.1.3 Muzzle velocity performance ballistic test. The selected sample cartridges shall be fired with an M67 propelling charge, zone 7.

4.4.2.2 Safety.

4.4.2.2.1 First article and initial production lot samples. Beginning with the first article and first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained in paragraphs 3.12.1-3.12.3, 15 sample projectiles shall be selected at random from each lot for this test. If any critical defect occurs, the lot shall be rejected.

4.4.2.2.2 Production lot samples. Once three consecutive lots have complied with 4.4.2.2.1, 5 sample projectiles shall be selected at random for this test. If any critical defect occurs, the lot shall be rejected. In the event that any lot fails to meet all the requirements of 3.12.1-3.12.3, successive lots must meet the provisions of 4.4.2.2.1 and the requirements specified therein.

4.4.2.2.3 Ballistic safety test. The selected sample cartridges shall be fired with an M67 propelling charge adjusted to produce 42,000 +/- 1,500PSI.

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4.4.3 Radiographic examination.

4.4.3.1 Determination of projectile explosive cast defects. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base at its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiographic image shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features shall project into the region of explosive under inspection, with the exception of specified Image Quality Indicators (IQIs) and radiation screening required to produce readable images in the thinner areas of the shell. Such screening shall be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with ASTM E1742 (see 6.4). In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.5 at every point within the projected area of explosive, including the base, with the maximum density no greater than that to permit adequate inspection over the entire area. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless positively identified as extraneous artifacts or metal parts irregularities. In the event an original radiographic image leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated 90 degrees about its longitudinal axis from the original exposure, and additional radiographic images taken for clarification. Acceptability in all cases shall be judged on the basis of the image with the largest projection, except in the case where a re-radiograph reveals multiple superimposed cavities, each of which individually complies with the requirements. All additional radiographic images of a projectile for defect conditions will be identified on the image as retakes and stored; traceable to the original.

4.4.3.2 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of Image Quality Indicators (IQI's), fabricated as specified herein. The three IQI's shall be placed on the radiographic source side of a projectile so that the 2T holes are at the point shown in Figure 2. When radiographing the frequency of the sensitivity check shall be no less than one in each fifty radiographic images, and on a rotating ring, the IQI's shall be used at one station for each complete rotation of the ring. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the IQI's, including the 2T holes, are perceptible in accordance with ASTM E 1742. The IQI's shall be made of steel having an approximate specific gravity of 7.8. The IQI's shall be located as shown on

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Figure 2. For further details on construction of the IQI's see ASTM E1742.

4.4.3.3 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurement on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Institute of Standards and Technology (NIST) or approved by the Government (see 6.4.1). The reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

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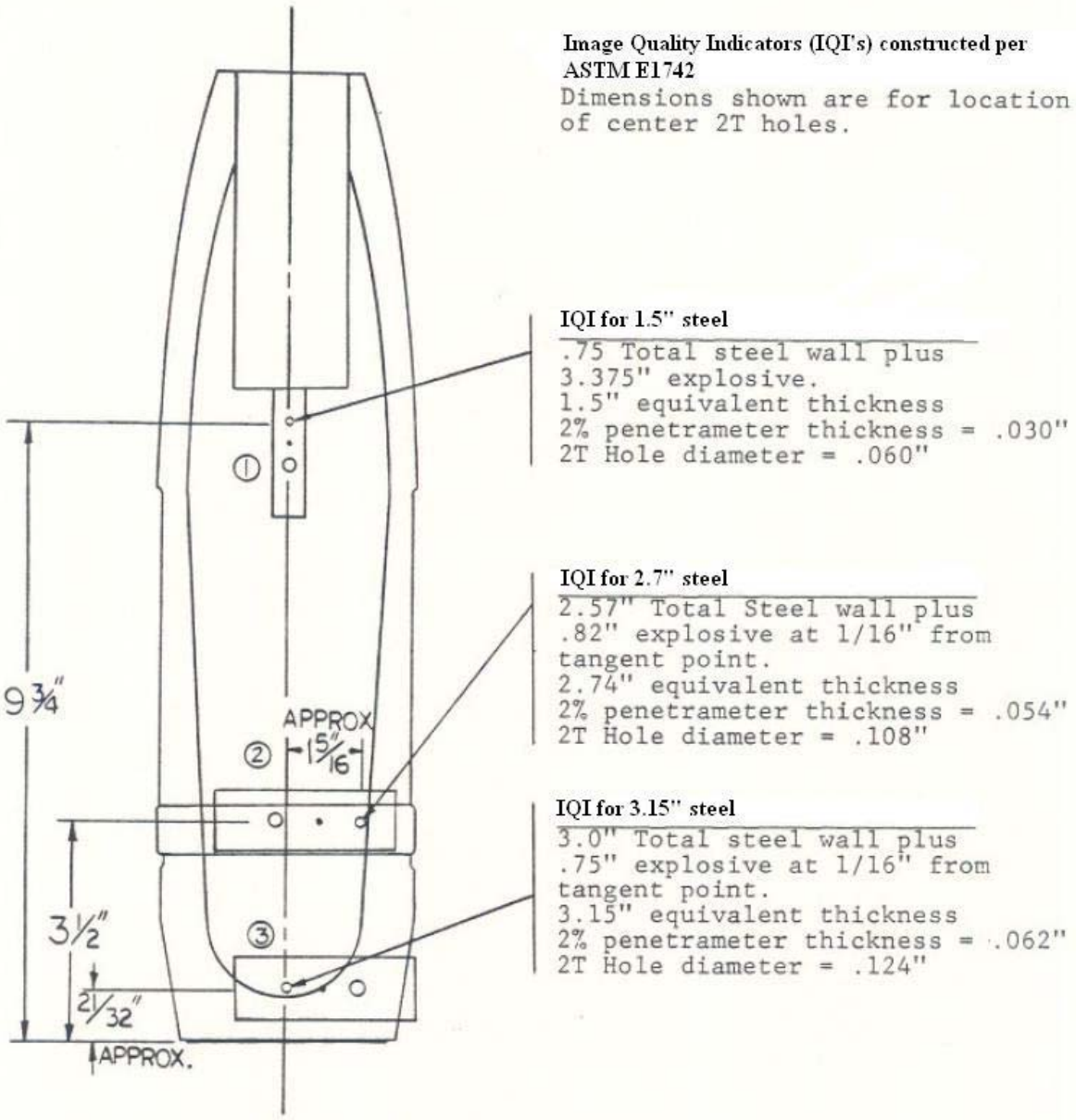


FIGURE 2. Projectile 105MM, HE, M1

Radiographic penetrameter requirements.

4.4.3.4 Review of radiographic sensitivity and techniques (see 6.4). The sensitivity of the radiographic technique shall be reviewed by submitting at least one radiographic image of the projectile containing the IQI and a copy of radiographic procedures prior to the start of initial production. These radiographic images shall be taken and processed using the same techniques as will be used in production. In addition one radiographic image of the IQI projectile and a sample of each type of rejectable defect shall be submitted for review once a month for the first three months of each contract and every

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three months thereafter. After qualification of the radiographic technique no changes to developing process, film, manufacturer or type radiographic source or exposure time shall be made without first submitting additional sample radiographic images of the IQI projectile for approval.

4.4.4 Check test for deterioration of primer, percussion, M28.

4.4.4.1 Head assembly non-functioning. The flash tube and the head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable. The separated primer heads shall be assembled into the test fixture (dwg. 8802117). A steel ball not weighing less than 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches). All head assemblies so tested shall be scrapped.

4.4.4.2 Head assembly functioning. This test shall be performed as specified in 4.4.4.1 except that the weight of the ball shall not be greater than 7.96 ounces, and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.4.4.3 Sample size. The Sampling plan for the functioning test shall be in accordance with Table IV. The Sampling plan for the non-functioning test shall be in accordance with Table V. Any head assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

TABLE IV. Functioning Test Samples

Sample	Sample size	Combined Sample Size	Acceptance Number	Rejection Number
First	20	20	*	2
Second	20	40	*	2
Third	20	60	0	2
Fourth	20	80	1	3
Fifth	20	100	1	3
Sixth	20	120	1	3
Seventh	20	140	2	3

* No acceptance on this sample

TABLE V. Non-functioning Test Samples

Sample	Sample size	Combined Sample Size	Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

* No acceptance on this sample

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for use on the 105MM, M119, M101 and M102 Howitzers. There is no known commercial use or application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. For acquisition purposes, the contract or order should specify packaging requirements. The projectiles should be palletized and marked in accordance with dwg 7549072 and be ordered by the applicable stock number.

c. Requirements for submission of first article sample.

d. Establishment of visual defect standards (see 6.3).

e. Submission of contractor Radiographic procedures (see 6.4).

f. Provisions for notification of metal parts defects (see 6.4.3).

g. Provisions for Certification of Nondestructive Test (NDT) personnel (see 6.5).

h. Qualification of post cyclic heat ovens (see 6.19).

i. Submission of contractor Acceptance Inspection Equipment (AIE)(see 6.7).

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j. Submission of ballistic test data and ammunition data cards (see 6.8).

k. Provisions for critical characteristic controls (see 6.9).

j. Proving ground safety test sampling plan (see 4.4.2).

6.3 Visual and X-ray image standards. Visual and X-ray image standards consistent with those used for metal parts acceptance will be established at or provided to the contractor's plant in compliance with 3.4.1, 3.4.4, 3.5.1 and 3.5.4.2. ARDEC Quality and Radiographic personnel will approve visual and X-ray standards.

6.4 Radiographic procedures. Submit radiographic procedures, standards and sample radiographic images to: Commander, ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, New Jersey 07806-5000.

6.4.1 Radiographic image density calibration. Densitometers used to perform radiographic image density measurements (4.4.3.3) should be verified in accordance with ASTM E-1079, Standard Practice for Calibration of Transmission Densitometers.

6.4.2 Radiographic film. Only a film type and vendor that has been approved by the Government should be used for radiographing this projectile. The final choice of film type must be determined for the particular radiographic facility to be used and should be based on achieving adequate sensitivity for both the IQI standard and inspected projectiles. A substitute item should not be used before prior testing and approval by the Government

6.4.2.1 Non-film methods of radiographic inspection. No non-film methods of radiographic inspection should be employed without prior review and approval by ARDEC. Qualification should be in accordance with ASTM E-1411 (Standard Practice for Qualification of Radioscopic Systems) and an approved qualification plan. System performance and implementation should comply with ASTM E-1255 (Standard Practice for Radioscopy). Qualification should include a demonstration that the specific non-film system proposed for production inspection detects or discloses all types and sizes of defects requiring radiographic examination.

6.4.3 Metal parts defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographic images. The rejected projectile will be held pending disposition by the procuring activity.

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6.5 Certification of NDT personnel. All personnel operating or calibrating NDT equipment should be qualified and certified in accordance with NAS-410 “Certification and Qualification of Nondestructive Test Personnel”. Application for copies should be addressed to the Aerospace Industries Association of America, Inc, (AIA/NAS), 1000 Wilson Blvd, Suite 1700 Arlington, VA 22209. Qualification to other standards must be submitted to and approved by Commander, ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000.

6.6 Projectile explosive contamination. Small amounts of explosive or exudates on the exterior surface of the projectile is unwanted but poses no safety hazard. Inspection for explosive or exudate residue on the threads or exterior of the projectile has always been a visual inspection. It is not feasible to place objective limits on a visual inspection for explosive contamination in the threads and on exterior surfaces. Some degree of interpretation is necessary to account for the subjectivity of the inspector and has been a part of the normal inspection operation at ammunition manufacturer plants for decades. The visual inspection requirement for explosive in the threads or on the exterior of the projectile is not intended to detect trace amounts of explosive. It is also not intended to use chemical reagents such as Webster’s to verify this requirement since most explosive items will test positive by nature of the loading process. Load plants have had responsibility for this requirement for many years and have performed safe assembly and delivery of numerous high explosive ammunition items.

6.6.1 Explosive Stain. A discoloration of the paint that has no visible explosive residue present. Stains are not acceptable at the LAP plant but are not a reason for non-acceptance at the Depot. See Figure 3.

6.6.2 Bloom. Explosive bloom (aka Frost, Needles, Dust et al) is a light fine crystalline particulate that can be present on the coatings or metal lattices of the steel projectile as a result of the melt pour loading process. Bloom represents a minute quantity of material and is not a safety hazard. Bloom will have a white "frosted" appearance and will sparkle when viewed with a light source in a dark environment. When wiped with a solvent bloom seems to come off easily with no visible residue but may reappear after the solvent has evaporated. Bloom is not a reason for non-acceptance at the Depot. See Figure 4.

6.6.3 Raw explosive. “cake” is tan or beige in color and is deposited on the projectile surface via a drip, spatter or other means. Raw caked explosive on the exterior of the projectile is not permissible anywhere in accordance with the LAP specification. Explosive cake will have a “bump” that can be felt as apposed to bloom that can not be felt. See Figure 5.

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6.7 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.8 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards should be forwarded to Commander, ARDEC, ATTN: AMSRD-AAR-AEM-C and AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000.

6.9 Critical characteristic controls. All production contracts should contain requirements for handling critical defects. As part of all proposals, suppliers should be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier’s integrated management system.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armaments, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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Figure 3. Projectile Stain

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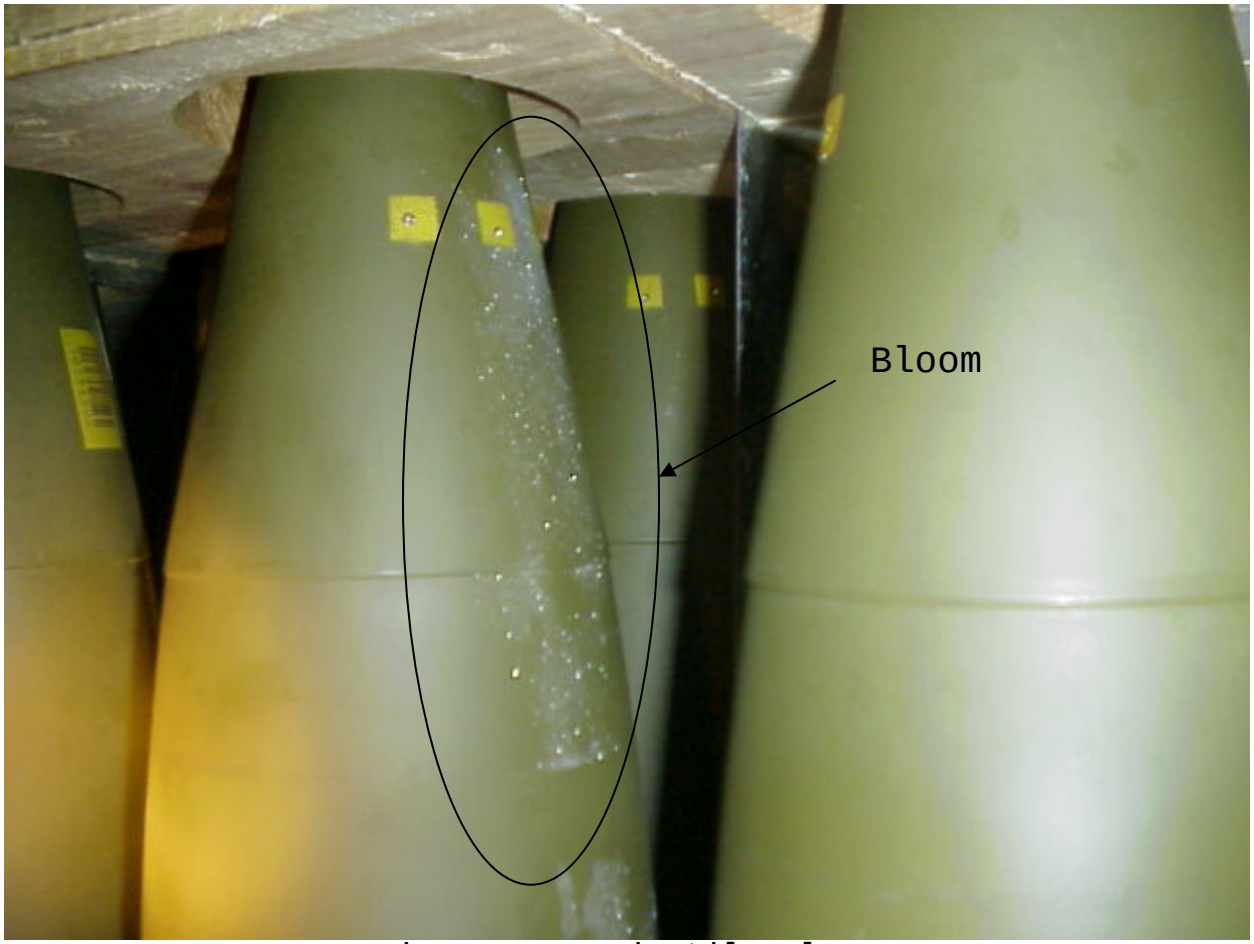


Figure 4. Projectile Bloom

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Figure 5. Caked Explosive

6.11 Protection of the rotating band. Care should be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

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6.12 Mating surfaces explosive contamination. If a chemical cleaner is required, acetone or methylene chloride should be used. Care should be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

6.13 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from Composition B risers, floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) should not be used (see 6.13.1).

6.13.1 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, risers and loading machines used in the pouring and cooling operations. Only TNT riser scrap is defined as clean.

6.13.1.1 Processing aid material for high explosive charge. Only qualified process aid materials are permitted with Composition B, MIL-C-401. Before adding materials, the Comp B temperature should be above 194 degrees Fahrenheit to provide satisfactory emulsion. The following material has been found acceptable as a wax dispersant when added to molten Comp B:

Lecithin, MIL-L-3061, in quantity of 0.10% - 0.02% by weight added.

6.14 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.15 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.16 Critical classification notes.

6.16.1 Base Separation, X-Ray. Base separation is the most serious explosive defect since it is exposed to the maximum loading during gun firing. The explosive loading can cause premature functioning either through adiabatic heating of entrapped air or by impact of the explosive with the projectile base across the gap. Although critical defects are defined as those “likely” to cause safety failures and the data available indicates that premature functioning is not likely for gaps up to .030 inches, experience with projectile loading and x-ray inspection indicates that a base separation of greater than .015

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inches must be classified as critical to maintain acceptably low escape rates for larger gaps. However in the case of the M1, the rounded base may cause difficulty in accurately reading the thickness if a given base separation. Thus, any visible separation is considered a major defect, while base separation in excess of .015" is still considered critical, as in the M107.

6.16.2 Cavities. Cavities in the bottom segment of the explosive cast over the sizes described are likely to cause premature functioning through adiabatic heating of entrapped air from loading during gun firing.

6.16.3 Annular Rings. Annular ring cavities in the bottom segments of the explosive cast over the width described are likely to cause premature functioning either through adiabatic heating of entrapped air or through explosive impact across the gap from loading during gun firing.

6.16.4 Piping Cavities. Piping cavities in the bottom segment of the explosive cast over the sizes described are likely to cause premature functioning through explosive collapse and impact across the void from loading during gun firing.

6.16.5 Cracks. Cracks in the bottom segment of the explosive cast over the width described are likely to cause premature functioning either through adiabatic heating of entrapped air or through explosive impact across the gap from loading during gun firing.

6.16.6 Explosive Cast Defects in Segment B. Explosive cast defects in Segment B in excess of twice the requirements are likely to cause premature functioning either through adiabatic heating of entrapped air, or through explosive impact across the voids

6.16.7 Metal Parts Defects. Metal parts defects, except rotating band gaps, large enough to be detected by x-ray examination of the explosive loaded projectile are likely to cause premature functioning from break-up of the projectile in the gun during firing.

6.16.8 Foreign Material, X-Ray. Foreign material in the explosive cast, detected by x-ray, is considered likely to cause a premature function because the effect of its infinite variety cannot be evaluated in advance of its occurrence.

6.16.9 Base Separation. Base separation in excess of .015 inches detected by physical measurement of sectioned sample projectiles is classified as critical because it indicates a problem with x-ray detection of base separation and/or the presence of a loose explosive cast, either of which represents a likely safety problem.

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6.16.10 Foreign Matter in the Charge Cavity. Foreign material in the charge cavity, detected by visual inspection prior to explosive filling, is considered likely to cause a premature function because its infinite variety cannot be evaluated in advance of its occurrence.

6.16.11 Supplementary Charge or pad Missing. When fired, a loose supplementary charge can impact the base of the fuze well with sufficient force to function the pressed TNT charge and the cast explosive projectile fill while in the gun tube. A missing supplementary charge can cause a safety failure if the fuze booster cup threads fail.

6.16.12 Projectile Zone Weight or Marking Incorrect. Incorrect projectile zone weight or zone weight marking can cause projectile impact short or long of the anticipated target area resulting in unacceptable collateral damage during combat operations in urban environments.

6.16.13 Ballistic Safety Test. Any premature projectile function or metal parts separation in the gun or in flight is likely to cause injury or death for the troops firing or supported by the howitzer.

6.16.14 Primer above flush. A primer above flush can result in early primer function when raising the breech or handling the cartridge. This is then likely to cause serious injury or death to those personnel handling the cartridge.

6.16.15 Incorrect sequence of increments. An incorrect sequence of increments may result in an incorrect propelling charge zone being fired. This can lead to a long or short round, which is likely to cause injury or death to Soldiers in the field.

6.17 Porosity.

6.17.1 Shrink porosity. Also known as core or sponge porosity, shrink porosity occurs during the cooling and solidification of the explosive cast, usually down the central axis of the cast, and radiating outward. Pores are often clustered and appear flattened or elongated.

6.17.2 Spherical porosity. Also known as gas porosity, the areas of spherical pores may occur anywhere throughout the explosive cast. Common causes are entrapped or entrained air induced during the melt pouring process. The X-ray image standards (see 6.3) for acceptable areas of spherical porosity should illustrate porous areas where the pores are generally spaced a minimum of two diameters apart in Segment A and the bottom half of segment B, and one diameter apart elsewhere in the cast. Sample loaded projectiles, sectioned to expose the porous areas, may be used to confirm pore spacing when setting

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standards. Pore spacing through and across an area of spherical porosity is usually equal.

6.18 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.3.2.1. Contractor inspection equipment designs should be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attribute data are preferred in order to facilitate the use of statistical process control.

6.19 Qualification of post-cyclic heat chambers. The chambers should be qualified for the number of projectiles and their location (stacking) in the chamber. The least efficient heat transfer area of the oven should be determined by thermocouples placed on projectiles within the chamber. An instrumented projectile (projectile with a thermocouple placed in the center both radially and axially of the explosive cast) should be placed in the least efficient area of the chamber and the required cycle time should be established based on these readings. The contractor should provide a qualification report. The report should contain a schematic of the chamber with the projectile layout with the supporting data confirming the least efficient are, a layout of chamber temperature distribution, a graph of the instrumented projectile results and a process plan. Forward results to Commander, ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ, 07806-5000.

6.20 Packaging Drawings. If the contractor chooses to build to the current approved TDP (7549073) the Government will waive the testing required to meet the First Article performance test requirements of the packaging specification; except for Performance Oriented Packaging (POP) testing which is still required to transport hazardous materials in accordance with Code of Federal Regulations (CFR), Title 49. Unless otherwise noted, the custodian of the following drawings is the U.S. Army Tank-Automated Command- Armament Research, Development and Engineering Center, Picatinny Arsenal, NJ 07806-5000;

6.21 Performance of primer check test. The test should be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges.

6.22 Ballistic test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test should be considered invalid and a new test should be performed with additional samples.

6.23 Ballistic test. The ballistic test (see 4.4.2) should be performed at a government proving ground.

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6.24 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.25 Subject term (key word) listing.
Base separation
Loading of High Explosive
Radiographic Inspection

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1315-2009-003)

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